Math Toolbox

$\sum_{k\ge0}a^k=\frac{1}{1-a}\ (|a|<1)$, $\sum_{k=0}^{N-1}a^k=\frac{1-a^N}{1-a}$, $\sum_{k\ge0}ka^k=\frac{a}{(1-a)^2}$. $z=a+jb=re^{j\theta}$, $e^{j\theta}=\cos\theta+j\sin\theta$, $\cos\theta=\frac{e^{j\theta}+e^{-j\theta}}{2}$, $\sin\theta=\frac{e^{j\theta}-e^{-j\theta}}{2j}$, $\frac{1}{z}=\frac{\bar z}{|z|^2}$; $|z_1z_2|=|z_1||z_2|$, $\angle z_1z_2=\angle z_1+\angle z_2$, $\angle\frac{z_1}{z_2}=\angle z_1-\angle z_2$. **Angle** from +Re axis, ccw: $\angle1=0$, $\angle j=\frac{\pi}{2}$, $\angle(-1)=\pi$, $\angle(1+j)=\frac{\pi}{4}$; 2nd quadrant ($a<0<b$): $\angle(a+jb)=\pi-\arctan\frac{b}{|a|}$. Real coeff. ⇒ complex roots in conj. pairs. $\left[\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right]^{-1}=\frac{1}{ad-bc}\left[\begin{smallmatrix}d&-b\\-c&a\end{smallmatrix}\right]$,
$\det(zI-A)=z^2-\mathrm{tr}A\,z+\det A$; triangular: eig. = diagonal.
$z^2-2\alpha z+c=0$: $z=\alpha\pm\sqrt{\alpha^2-c}$; complex ⇒ $|z|=\sqrt{c}$.
$1\pm e^{-j\Omega}=e^{-j\Omega/2}(e^{j\Omega/2}\pm e^{-j\Omega/2})$:
$1-e^{-j\Omega}=2j\sin\frac{\Omega}{2}e^{-j\Omega/2}$, $1+e^{-j\Omega}=2\cos\frac{\Omega}{2}e^{-j\Omega/2}$.

W1 Signals

CT $x(t)$, $t\in\mathbb{R}$; DT $x[n]$, $n\in\mathbb{Z}$ (sequence); $\{x[n]\}$ whole signal. Sequence notation: ↑ marks $n=0$, unlisted samples $=0$ (count from the arrow).
$\alpha x[n]$ scaling; $x[n-k]$ ($k>0$) **delay** = shift **right**; $x[n+k]$ **advance** = shift **left**; $x[-n]$ flip about $n=0$ ($x[0]$ stays). Locate a sample: set argument = its index. **Combined** (read inside out): $x[-n+k]=x[-(n-k)]$: flip, then shift right by k; value at n is $x[k-n]$.
$\delta[n]=1$ iff $n=0$ (else 0); $s[n]=1$ iff $n\ge0$.
$\boxed{s[n]=\sum_{k\le n}\delta[k],\ \delta[n]=s[n]-s[n-1]}$

Impulse decomposition $\boxed{x[n]=\sum_k x[k]\,\delta[n-k]}$ (one term per nonzero sample). Sifting: $x[n]\delta[n-k]=x[k]\delta[n-k]$. **Causal signal**: $x[n]=0\ \forall n<0$. **Bounded**: $|x[n]|\le M<\infty\ \forall n$.

W1 System Properties

System G: $u\mapsto y=Gu$. A property holds for *all* inputs: prove with the definition, disprove with *one* counterexample.

Memoryless	$y[n]$ depends only on $u[n]$ (any other index ⇒ memory)
Causal	only on $u[k]$, $k\le n$ (index $>n$, or $u[-n]$ ⇒ non-causal)
Linear	$G\{\alpha_1u_1+\alpha_2u_2\}=\alpha_1Gu_1+\alpha_2Gu_2\ \forall u_i,\alpha_i$; implies $0\mapsto0$
Time-inv.	$u_1[n-k]\mapsto y_1[n-k]\ \forall k$: compute $G\{u_1[n-k]\}$, compare with $y_1[n-k]$ (replace *every* n by $n-k$)
BIBO stable	$|u[n]|\le1\ \forall n\Rightarrow|y[n]|\le M<\infty$; disprove with one bounded u giving unbounded y (try $u=s$)

Linearity fails: constant offset (affine ≠ linear), u^2, $|u|$, $\cos u$, products $u[n]u[n-1]$. **TI fails:** explicit n in coefficients ($nu[n]$, $g[n]u[n]$), time scaling $u[2n]$, reversal $u[-n]$ (delayed input comes out advanced). $g[n]u[n]$ is linear for every fixed g. Accumulator $\sum_{k\le n}u[k]$ and $y[n]=ay[n-1]+u[n]$, $|a|\ge1$: unstable. Bounded nonlinearities ($\cos u$) are stable. Linear but not TI: response to δ does not give response to shifted inputs. Physical systems are causal; non-causal = offline processing.

W2 Impulse Response & Convolution

$\boxed{h[n]:=G\{\delta[n]\}}$. LTI shift rule: $\delta[n-k]\mapsto h[n-k]$ (TI), $c\delta[n-k]\mapsto ch[n-k]$ (L). For LTI, h determines the output for every input; not for non-LTI.

$$\boxed{y[n]=(u*h)[n]=\sum_k u[k]\,h[n-k]=\sum_k h[k]\,u[n-k]}$$

Needs L *and* TI. Flip & slide (either signal may be flipped), or $y[n]=\sum_{i+j=n}u[i]h[j]$ (polynomial product). **Length** $L_y=L_u+L_h-1$; first index $=n_{u,0}+n_{h,0}$, last = sum of last indices. Check: $\sum_n y=\sum u\cdot\sum h$.
Properties: commutative, associative, distributive; identity $u*\delta=u$; $u*\delta[n-k]=u[n-k]$; ! $u*h\ne u\cdot h$. **Series** ⇒ h_1*h_2 (LTI blocks may be interchanged; not for non-LTI), **parallel** ⇒ h_1+h_2, direct path ⇒ $+\delta$. **Deconvolution** ($h[0]\ne0$):
$u[n]=(y[n]-\sum_{k\ge1}h[k]u[n-k])/h[0]$ recursively.
Step response
$\boxed{r[n]=(h*s)[n]=\sum_{k\le n}h[k]}$, $\boxed{h[n]=r[n]-r[n-1]}$ (either one characterizes the LTI system). Stable: $r[\infty]=\sum_k h[k]=H(1)$.

Causality, Stability, FIR/IIR (LTI)

$\boxed{\text{causal}\iff h[n]=0\ \forall n<0}$ (nonzero $h[m]$, $m<0$, weighs the future input $u[n-m]$). Causal *system* ≠ causal *signal*. Memoryless $\iff h=c\delta$. $\boxed{\text{BIBO stable}\iff\sum_k|h[k]|<\infty}$, since $|y[n]|\le\sum_k|h[k]|\max|u|$. $\alpha^n s[n]$: $\sum=\frac{1}{1-|\alpha|}$, stable iff $|\alpha|<1$. ! Traps: causal ⇏ stable; bounded h ⇏ stable ($s[n]$); $h\to0$ ⇏ stable ($\frac{1}{n+1}s[n]$, harmonic sum); left-sided a^n ($n\le0$) with $|a|>1$ is stable (support decides, not base); an unstable block can be cancelled in a cascade ($s*(\delta-\delta[n-1])=\delta$): judge the overall h. **FIR**: finitely many $h[n]\ne0$ ⇒ always stable. **IIR**: infinitely many (stable or not). Feedback $y[n]=u[n]+\alpha y[n-D]$: $h=\sum_{k\ge0}\alpha^k\delta[n-kD]$, IIR, stable iff $|\alpha|<1$.

W3 System Representations

$h[n]$ (analysis) · LCCDE (real-time/DSP) · state space (simulation, design). Fixed-memory real time: recursion & SS yes; convolution with IIR h no.

LCCDE & State Space

$\boxed{\sum_{k=0}^{N}a_k y[n-k]=\sum_{k=0}^{M}b_k u[n-k]}$, $a_0\ne0$, LTI by definition, order N = largest output delay. Recursion: $y[n]=\frac{1}{a_0}(\sum_{k=0}^{M}b_k u[n-k]-\sum_{k=1}^{N}a_k y[n-k])$. **LCCDE→$h$:** iterate with $u=\delta$, $y[n]=0$ for $n<0$. ! $y[n]-ay[n-1]$ has ratio $+a$. h → **LCCDE:** FIR: $b_k=h[k]$; sum of geometric terms: via $H(z)$.

$$\boxed{q[n+1]=Aq[n]+Bu[n],\quad y[n]=Cq[n]+Du[n]}$$

$A_{N\times N}$, $B_{N\times1}$, $C_{1\times N}$, D = feedthrough (instant $u\to y$). **State** = everything from the past that still affects the future: equal $q[n]$ + equal future input ⇒ equal future output. Dimension N = order; inputs are not states; one state per stored value / energy store.
SS→h ($q[0]=0$, $q[n]=A^{n-1}B$):
$\boxed{h=\{\underset{\uparrow}{D},CB,CAB,CA^2B,\dots\}}$, $h[n]=CA^{n-1}B$ ($n\ge1$).
! $h[0]=D$, not CB; $D=0\Rightarrow$ response 1 sample late. Initial state adds free response $CA^nq[0]$.
LCCDE→SS (companion form, $a_0=1$, RHS $b_0u[n]$, $q=[y[n-N],..,y[n-1]]^\top$): $A=\left[\begin{smallmatrix}0&1&0\\0&0&1\\-a_3&-a_2&-a_1\end{smallmatrix}\right]$, $B=\left[\begin{smallmatrix}0\\0\\b_0\end{smallmatrix}\right]$ (shown $N=3$), $C=[-a_N\cdots-a_1]$, $D=b_0$. Delayed inputs ($M\le N$): same A, $B=[0..01]^\top$, $C=[b_N-a_Nb_0,..,b_1-a_1b_0]$, $D=b_0$; or put past inputs into the state.
SS→LCCDE: eliminate q (equation one step earlier, subtract) or via $H(z)$; scalar: $y[n]-Ay[n-1]=Du[n]+(CB-AD)u[n-1]$.
Realizations not unique: $q'=Tq$: (TAT^{-1},TB,CT^{-1},D) gives the same h. Compare realizations via h or $H(z)$, never entrywise.

CT State Space & Linearization

$\boxed{\dot q=A_cq+B_cu,\ y=C_cq+D_cu}$. Build: rows $\dot q_1=q_2$ (definitions), last row = physical law solved for the highest derivative. $m\ddot p+c\dot p+kp=F$, $q=[p,\dot p]^\top$: $A_c=\left[\begin{smallmatrix}0&1\\-\frac{k}{m}&-\frac{c}{m}\end{smallmatrix}\right]$, $B_c=\left[\begin{smallmatrix}0\\\frac{1}{m}\end{smallmatrix}\right]$.

1. Equilibrium (q_e,u_e): $\boxed{f(q_e,u_e)=0}$, $y_e=h(q_e,u_e)$; $u_e\ne0$ allowed, several may exist, $\partial f/\partial q=0$ not required. **2.** Jacobians evaluated at (q_e,u_e) (row = equation, column = variable): $A_c=\frac{\partial f}{\partial q}|_e$, $B_c=\frac{\partial f}{\partial u}|_e$, $C_c=\frac{\partial h}{\partial q}|_e$, $D_c=\frac{\partial h}{\partial u}|_e$. **3.**
$\boxed{\Delta\dot q=A_c\Delta q+B_c\Delta u,\ \Delta y=C_c\Delta q+D_c\Delta u}$, $\Delta q=q-q_e$, $\Delta u=u-u_e$, $\Delta y=y-y_e$. Constant matrices ⇒ LTI; valid only for small deviations (neglected higher-order terms grow faster). ! No q, $\sqrt{q}$, $\sin q$ left in the matrices. One linear model per equilibrium.

W4 Sampling & Discretization

Sampler S (CT→DT): $\boxed{x[n]=x(nT_s)}$, $f_s=\frac{1}{T_s}$; values between samples lost. **ZOH** H (DT→CT): $\boxed{x(t)=x[n],\ t\in[nT_s,(n+1)T_s)}$, staircase (linear interpolation would be first-order hold). $S\circ H=\mathrm{id}$, $H\circ S\ne\mathrm{id}$.

$$u[n]\to\boxed{H}\to u(t)\to\boxed{G_c}\to y(t)\to\boxed{S}\to y[n]\qquad G_d=SG_cH$$

Composition read right to left; $u[n],y[n]$ DT, $u(t),y(t)$ CT; under ZOH $u(t)$ constant per interval (no averaging of $u[n],u[n+1]$). Pure delay $kT_s<\tau\le(k+1)T_s\Rightarrow y[n]=u[n-k-1]$ (rounded up).

Euler & Exact Discretization

Euler $\dot q\approx\frac{q(t+T_s)-q(t)}{T_s}$:
$\boxed{A_d=I+T_sA_c,\ B_d=T_sB_c,\ C_d=C_c,\ D_d=D_c}$; output eq. has no derivative ⇒ C,D unchanged. Error $\mathcal{O}(T_s^2)$/step. Stable iff $|1+T_s\lambda_i(A_c)|<1$; scalar $a<0$: $\boxed{T_s<2/|a|}$, monotone iff $T_s<1/|a|$; for $\frac{1}{|a|}<T_s<\frac{2}{|a|}$ decays with sign flips; beyond: growing alternation (stable CT model becomes unstable).
Exact (ZOH): input held, augment $r=[q;u]$, $\dot r=Mr$: $\boxed{M=\left[\begin{smallmatrix}A_c&B_c\\0&0\end{smallmatrix}\right]}$, $e^{MT_s}=\left[\begin{smallmatrix}A_d&B_d\\0&I\end{smallmatrix}\right]$, $A_d=e^{A_cT_s}$ (top-left), B_d top-right ($=A_c^{-1}(A_d-I)B_c$), $C_d=C_c$, $D_d=D_c$. Scalar: $\boxed{a_d=e^{aT_s},\ b_d=\frac{b}{a}(e^{aT_s}-1)}$ ($a=0$: $b_d=bT_s$);
$M^k=\left[\begin{smallmatrix}a^k&a^{k-1}b\\0&0\end{smallmatrix}\right]$. Eigenvalues $\lambda\mapsto e^{\lambda T_s}$, $|e^{\lambda T_s}|=e^{\mathrm{Re}\lambda T_s}$: stability kept for every T_s; stable 1st order $a_d\in(0,1)$. Exact at samples, inter-sample behaviour lost. Euler = exponential series truncated after linear term.

	A_d	B_d	accuracy / stability
Euler	$I+T_sA_c$	T_sB_c	$\mathcal{O}(T_s^2)$; stable only for small T_s
Exact	$e^{A_cT_s}$	top-right of e^{MT_s}	exact at samples; always kept

Fast dynamics / large T_s ⇒ exact; slow dynamics & small T_s ⇒ Euler OK.
$e^{At}=\sum\frac{(At)^k}{k!}$ (*not* entrywise): diag ↦ diag(e^{λ_it}); nilpotent: series ends; $\left[\begin{smallmatrix}\lambda&1\\0&\lambda\end{smallmatrix}\right]\mapsto e^{\lambda t}\left[\begin{smallmatrix}1&t\\0&1\end{smallmatrix}\right]$; $\left[\begin{smallmatrix}\sigma&\omega\\-\omega&\sigma\end{smallmatrix}\right]\mapsto e^{\sigma t}\left[\begin{smallmatrix}\cos\omega t&\sin\omega t\\-\sin\omega t&\cos\omega t\end{smallmatrix}\right]$; $A=V\Lambda V^{-1}\Rightarrow Ve^{\Lambda t}V^{-1}$. Integrator $\dot q=u$: $(1,T_s)$; double integrator $\dot p=v,\dot v=u$: $A_d=\left[\begin{smallmatrix}1&T_s\\0&1\end{smallmatrix}\right]$, $B_d=\left[\begin{smallmatrix}T_s^2/2\\T_s\end{smallmatrix}\right]$ (Euler misses $T_s^2/2$).

W5 The z-Transform & z-Plane

z_0 (number) → z_0^n (signal) → $H(z_0)$ (eigenvalue, one number) → $H(z)$ (function on ROC). **Eigenfunction:**
$\boxed{u=z_0^n\ \forall n\Rightarrow y=H(z_0)z_0^n}$, $H(z_0)=\sum_k h[k]z_0^{-k}$ (z_0 in ROC).
$u\equiv1\Rightarrow y=H(1)$ (DC gain). $z^n=r^ne^{j\theta n}$: r envelope (<1 decay, $=1$ sustained, >1 growth), θ phase advance per sample.

$$\boxed{X(z)=\sum_{n=-\infty}^{\infty}x[n]z^{-n}},\quad Y(z)=H(z)U(z)$$

ROC: never contains poles; causal/right-sided $|z|>\max|p|$, left-sided $|z|<\min|p|$, two-sided annulus, causal FIR all $z\ne0$. Expression unique only with ROC or causality ($a^ns[n]$ and $-a^ns[-n-1]$ both $\frac{z}{z-a}$). **Any LTI: stable ⟺ ROC ∋ unit circle.** Delays keep the ROC (except $c\delta$ loses $z=0$).

$x[n-k]$	$z^{-k}X$	$a^nx[n]$	$X(z/a)$
$x[n+k]$	z^kX	$nx[n]$	$-zX'(z)$
$u*h$	UH	$x[-n]$	$X(1/z)$
$\delta[n-k]$	z^{-k}	$\sum_{k\le n}x[k]$	$\frac{z}{z-1}X$
$a^ns[n]$	$\frac{z}{z-a}$, $\lvert z\rvert>\lvert a\rvert$	$s[n]$	$\frac{z}{z-1}$
$a^{n-1}s[n-1]$	$\frac{1}{z-a}$	$na^ns[n]$	$\frac{az}{(z-a)^2}$
$(n+1)a^ns[n]$	$\frac{z^2}{(z-a)^2}$	$ns[n]$	$\frac{z}{(z-1)^2}$

$r^n\cos(\theta n)s[n]\leftrightarrow\frac{z(z-r\cos\theta)}{z^2-2r\cos\theta z+r^2}$,
$r^n\sin(\theta n)s[n]\leftrightarrow\frac{r\sin\theta z}{z^2-2r\cos\theta z+r^2}$. IVT $x[0]=\lim_{z\to\infty}X$ (causal); FVT $x[\infty]=\lim_{z\to1}(z-1)X$ (if limit exists). FIR: $H(z)=\sum_k h[k]z^{-k}$, all poles at 0. A zero on the unit circle at angle θ annihilates $e^{j\theta n}$ (zero at 1: constants, at -1: $(-1)^n$).

Obtaining $H(z)$

$\boxed{H(z)=\frac{b_0+b_1z^{-1}+\cdots+b_Mz^{-M}}{1+a_1z^{-1}+\cdots+a_Nz^{-N}}}$ (zero IC, delay → z^{-k}), then multiply by $z^{\max(N,M)}$. ! Move all y-terms to the left first (signs flip). $\boxed{H(z)=C(zI-A)^{-1}B+D}$, $(zI-A)^{-1}=\frac{\mathrm{adj}(zI-A)}{\det(zI-A)}$: poles $\subseteq$ eig(A) (an eigenvalue may cancel with a zero). Feedback $u=-Kq+d$: $A_K=A-BK$.

Poles, Modes, Partial Fractions

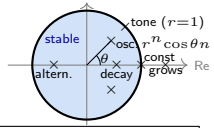
$H=b_0z^{N-M}\frac{\prod(z-z_i)}{\prod(z-p_k)}$: zeros ○ (numerator roots), poles × (denominator roots); $N>M$: zeros at 0. **PFE:** $\frac{H}{z}=\frac{d}{z}+\sum\frac{c_k}{z-p_k}$, $c_k=[(z-p_k)\frac{H}{z}]_{z=p_k}$, $d=H(0)$:

$$\boxed{H=d+\sum_k\frac{c_kz}{z-p_k}\Rightarrow h[n]=d\delta[n]+\sum_k c_kp_k^ns[n]}$$

One mode p_k^n per pole, weights c_k from the numerator. Conj. pair: $cp^n+\bar c\bar p^n=2|c|r^n\cos(\theta n+\angle c)$. Double pole: $(n+1)p^n$ (input at a pole: bounded if $|p|<1$). Settling to a constant ≠ stable (pole at 1).

Stability & Pole Placement

$\boxed{\text{causal LTI stable}\iff\text{all poles }|p_k|<1}$; SS: all $|\lambda(A)|<1$ suffices. Zeros irrelevant; coefficients ≠ poles (a coefficient >1 says nothing). 2nd order $z^2+a_1z+a_0$ stable $\iff|a_0|<1,|a_1|<1+a_0$. $0<p<1$: smooth decay, never changes sign; $-1<p<0$: alternating decay; $re^{\pm j\theta}$, $r<1$: decaying oscillation; $|p|=1$: sustained (tone), $p=1$ constant persists (not BIBO); $|p|>1$: grows. Two real modes: fast one dies first.
Dominant mode = largest $|p|$.



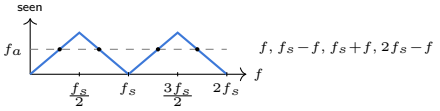
$\boxed{(z-re^{j\theta})(z-re^{-j\theta})=z^2-2r\cos\theta z+r^2}\Rightarrow$
$y[n]=2r\cos\theta\,y[n-1]-r^2y[n-2]+u[n]$ (numerator z^2). Period N_p samples: $\boxed{\theta=2\pi/N_p}$; envelope $\to\varepsilon$ after N samples: $\boxed{r=\varepsilon^{1/N}}$; in Hz: $f=\theta f_s/2\pi$. Placement is in samples: changing f_s for the same physical response changes θ and r ($f_s/2$: $\theta\to2\theta$, $r\to r^2$). CT pole λ ↔ DT pole $e^{\lambda T_s}$.

W6 Frequency in DT & Sampling

Ω [rad/sample] = phase advance per sample; one cycle $=2\pi/\Omega$ samples; $\cos\Omega n=\mathrm{Re}\,e^{j\Omega n}$. $\boxed{\Omega=\frac{2\pi f}{f_s}=\omega T_s,\ f=\frac{\Omega f_s}{2\pi}}$; $\frac{f}{f_s}=$ periods per sample. Ω alone has no Hz (needs f_s).
$e^{j(\Omega+2\pi)n}=e^{j\Omega n}\Rightarrow$ every distinct frequency in $\boxed{\Omega\in(-\pi,\pi]}$ (axis is a circle). $\Omega=0$ constant; $\Omega=\pi$: $(-1)^n$, fastest = **Nyquist frequency** $f_s/2$. ! Reduce Ω mod 2π into $(-\pi,\pi]$ before comparing speeds; closest to π is fastest.
Periodic $\iff\boxed{\Omega/2\pi=k/N\in\mathbb{Q}}$; period N for k/N irreducible (DT period is an integer, not $2\pi/\Omega$). Sum of periodic signals: period $=\mathrm{lcm}$ of periods.
Decomposition/spectrum: $x[n]=\sum_i A_i\cos(\Omega_in+\phi_i)$, $A\cos(\Omega n+\phi)=\frac{A}{2}e^{j(\Omega n+\phi}+\frac{A}{2}e^{-j(\Omega n+\phi)}$: two lines at $\pm\Omega_i$, magnitude $\frac{A_i}{2}$ each; constant c: one line at 0 of height $|c|$; $A\cos\pi n$: one line at π of height $|A|$ ($\pm\pi$ coincide). Spectra add (no averaging); magnitude spectrum hides phase (sign = phase shift π).

Aliasing, Folding, Nyquist

Aliases: $\boxed{\pm f+kf_s,\ k\in\mathbb{Z}}$ give identical samples ($f\pm kf_s$ alone misses the mirror branch). Branch $f+kf_s$: same phase; branch $-f+kf_s$: opposite phase (sine → −sine). **Folding** (alias seen in $[0,\frac{f_s}{2}]$): (1) subtract multiples of f_s into $(-\frac{f_s}{2},\frac{f_s}{2}]$; (2) drop the sign. $\boxed{f_a=|f-f_s\,\mathrm{round}(f/f_s)|}$. ! Fold with f_s, not $f_s/2$. A frequency sweep is heard as a triangle:



Nyquist: samples determine the signal uniquely $\iff\boxed{f_s>2f_{\max}}$ (strict: $\sin$ at exactly $f_s/2$ gives all-zero samples). Content above $f_s/2$ is *not* removed: it folds onto genuine content as false tones; aliasing cannot be undone after sampling (a digital filter would also remove the genuine content there). **Anti-aliasing filter**: CT low-pass *before* the sampler, cutoff $f_c\approx f_{\max}$ of interest; real filters roll off gradually ⇒ place $f_s/2$ above the transition band (margin). Disturbances *inside* the band (e.g. 50 Hz hum below f_c) pass and are not aliased: need a separate filter; raising f_s does not help them. Choosing f_s: $f_{\max}\to f_c\to f_s>2f_c$ + margin.

W7 DTFT & Frequency Response

$$\boxed{X(\Omega):=X(z)\big|_{z=e^{j\Omega}}=\sum_{n=-\infty}^{\infty}x[n]\,e^{-j\Omega n}}$$

DTFT = z-transform on the unit circle; frequency Ω = angle of the point $e^{j\Omega}$: $\Omega=0$ DC ($z=1$), $\frac{\pi}{2}$ ($z=j$), π Nyquist ($z=-1$). Of a signal: **spectrum**; of h: **frequency response** $H(\Omega)$. Complex: $X(\Omega)=|X(\Omega)|e^{j\angle X(\Omega)}$ (magnitude, phase). Periodic in Ω with period 2π.
Existence: sum converges for every absolutely summable x (every finite-length signal). Sinusoids/steps: not summable, DTFT only in a generalized sense. Causal rational H:

$\boxed{\text{unit circle} \subset \text{ROC} \iff \text{poles inside} \iff \text{stable} \iff \sum |h| < \infty}$
frequency response exists iff stable; unstable systems (poles on/outside circle, e.g. $s[n]$, $(-1)^n$, $2^n s[n]$) have none.
Real signals: $\boxed{X(-\Omega) = X(\Omega)^*}$: $|X|$ even, $\angle X$ odd $\Rightarrow$ plots over $[0, \pi]$ suffice (same for real h).

Computing & Geometric Evaluation

At $z = e^{j\Omega}$ evaluate numerator and denominator (two complex numbers): $\boxed{|H(\Omega)| = \frac{|\text{num}|}{|\text{den}|},\ \angle H(\Omega) = \angle\text{num} - \angle\text{den}}$. On a zero: $H{=}0$, phase undefined.

$$\boxed{|H(\Omega)| = |b_0| \frac{\prod_i |e^{j\Omega} - z_i|}{\prod_k |e^{j\Omega} - p_k|} = |b_0| \frac{\prod \text{dist. to zeros}}{\prod \text{dist. to poles}}}$$

$\angle H(\Omega) = \angle b_0 + \sum \angle(e^{j\Omega} - z_i) - \sum \angle(e^{j\Omega} - p_k)$. Poles/zeros at the origin: distance 1, affect phase only.

- **Zero on the unit circle** at angle θ: $|H(\theta)| = 0$ (notch, frequency removed).
- **Pole near the unit circle**: peak near its angle; closer $\Rightarrow$ higher and narrower.
- **Pole close to a zero**: distances nearly equal $\Rightarrow$ $|H|$ almost constant away from the pair.
- Near a pole large, on a zero vanishes; evaluation point moving away from the poles $\Rightarrow$ $|H|$ decreases.
- 1st order $H = \frac{z}{z-a} = \frac{1}{1-ae^{-j\Omega}}$: $|H(0)| = \frac{1}{|1-a|}$, $|H(\pi)| = \frac{1}{|1+a|}$; $a{>}0$ low-pass, $a{<}0$ high-pass; $|H|^2 = \frac{1}{1-2a\cos\Omega+a^2}$.
- Delay z^{-k}: $H = e^{-jk\Omega}$, $|H| = 1$ (0 dB), linear phase $-k\Omega$. First difference $1{-}z^{-1}$: $H = 2\sin\frac{\Omega}{2}\, e^{j(\frac{\pi}{2} - \frac{\Omega}{2})}$ (high-pass, $H(0){=}0$, linear phase). $1{+}z^{-1}$: $2\cos\frac{\Omega}{2} e^{-j\Omega/2}$ (low-pass, $H(\pi){=}0$).

Bode Plot

Over $\Omega \in [0, \pi]$: magnitude $\boxed{20\log_{10}|H(\Omega)|}$ [dB] and phase $\angle H(\Omega)$ [rad]. Compute H at selected frequencies (geometrically or from the formula), convert, connect. $|H|$: $10 \to +20$, $2 \to +6$, $\sqrt{2} \to +3$, $1 \to 0$, $0.707 \to -3$, $0.5 \to -6$, $0.1 \to -20$, $0.01 \to -40$, $0 \to -\infty$ dB; $|H| = 10^{\text{dB}/20}$; every -20 dB divides by 10; products of factors $\Rightarrow$ dB add.

Response to Sinusoids

Stable LTI, $u[n] = e^{j\Omega_0 n}$ for all n: $\boxed{y[n] = H(\Omega_0)\, e^{j\Omega_0 n}}$ (eigenfunction, one complex multiplication). Real h: $\boxed{u = A\cos(\Omega_0 n + \phi) \Rightarrow y = A|H(\Omega_0)|\cos(\Omega_0 n + \phi + \angle H(\Omega_0))}$: same frequency, amplitude scaled by $|H(\Omega_0)|$, phase shifted by $\angle H(\Omega_0)$ (delay of $-\angle H/\Omega_0$ samples). Sums of sinusoids: apply per component (superposition). Constant input c: $y = cH(0)$.
Steady state vs. transient: for $u = \cos(\Omega_0 n)s[n]$ (switched on at $n{=}0$) and a stable system: $y[n] \to |H(\Omega_0)|\cos(\Omega_0 n + \angle H(\Omega_0))$ as $n \to \infty$; the difference (transient) consists of the decaying modes p_k^n. Formula exact only if input applied for all n. A system that creates new frequencies (harmonics, distortion) is *not* LTI; LTI systems only scale and shift each frequency.